

Web Technology

Beginning JavaScript

Course Outline

Lecture 7:

7-1 Introduction to JavaScript

7-2 Usage of Data Types and Variables

7-3 Basic Data Calculation and String Manipulation

Lesson Information

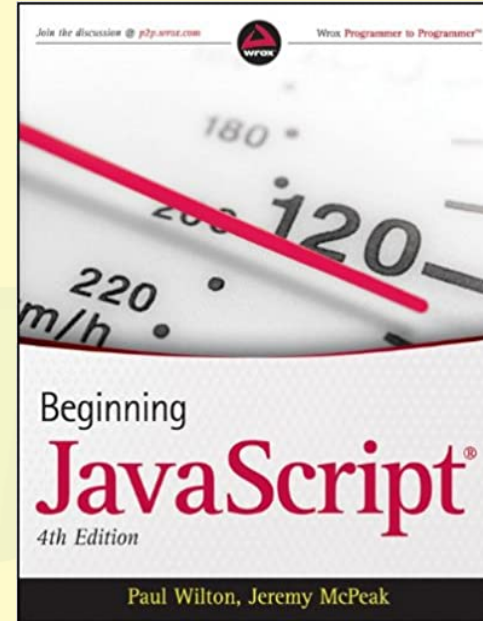


UNIVERSITY
of
INFORMATION TECHNOLOGY

**TEXT
BOOK**



<https://bbooks.info/book/beginning-javascript-4th-edition>



Lesson Information



prerequisites

- ★ Basic computer literacy
- ★ a basic understanding of **HTML** and **CSS**



- ◆ To understand the principles of creating an effective web page and to familiar with what JavaScript is,
- ◆ To develop basic programming skills using JavaScript
- ◆ To discuss how it fits into a web site

7-1 Introduction to JavaScript

What Is JavaScript?

- programming language designed for Web pages
- used primarily to create a dynamic and interactive for the user
- executed only when an event occurs(Mouse movement , Key press Page load, etc.)
- runs on the client (Web browser)
- scripting language (no compilation required)

Why Choose JavaScript?

- ★ main reason is its widespread use and availability
- ★ most commonly used browsers, IE and Firefox support JavaScript



How to put JavaScript code into web page

- ✎ Inserting JavaScript into a web page is much like inserting any other HTML content;
- ✎ Use tags to mark the start and end of your script code. The tag used to do this is **<script>**.
- ✎ This tells the browser that the following chunk of text, the bounded by the closing **</script>** tag, is not HTML to be displayed, but rather script code to be processed.
- ✎ The chunk of code surrounded by the **<script>** and **</script>** tags is called a **script block**.





Where should insert JavaScript code into web

can put the `<script>` tags between the `<head>` and `</head>` tags, or

can between the `<body>` and `</body>` tags

before the `<html>` tag or after the `</html>` tag —is **not permitted** in the web standards and considered bad practice.

```
<html>
<head>.....</head>
<body >
.....
</body>
</html>
```


Script Attribute

- ★ The `<script>` tag has a number of attributes, but the most important one is **type**.
- ★ different scripting languages are available, so need to tell the browser which scripting language is used.
opening script tag will look like this:
- ★ `<script type="text/javascript">`



3 ways to attach JavaScript program in a Web page



URI to a separate file

```
<script type="text/javascript"  
src="/js/external.js">  
</script>
```



write the script directly into
the HTML file within the
<script> tag

```
<script type="text/javascript">  
.....JavaScript Code.....  
</script>
```



write the script directly
within an HTML element

```
<p onmouseover="alert('WoW') ">  
Move mouse here!</p>
```

Good Usage of JavaScript

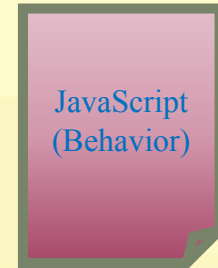


Bad(All in the same file)



Not Recommended

Good(Separate files)



Recommended

Typical Programming Process

```
<html>
<head>
</head>
<body >
<h3 onmouseover="test();">Move mouse here</h3>
<hr>
<p id="P1">COLOR ME!</p>
</body>
<script type="text/javascript">
function test()
{
    document.getElementById("P1").style.color="red";
}
</script>
</html>
```

Associate an event to a JavaScript function

Give an id to the element that you want to manipulate

Write a function to be executed

Refer to the target element by its ID

JavaScript Comments

- can be used to explain JavaScript code, and to make it more readable.
- can also be used to prevent execution, when testing alternative code.

Single Line Comments

Single line comments start with `//`.
Any text between `//` and the end of the line will be ignored by JavaScript (will not be executed).

`// It is single line comment`

```
document.write("hello javascript");
```

JavaScript Comments

Multi-line comments

start with `/*` and end with `*/`.
Any text between `/*` and `*/`
will be ignored by JavaScript
(will not be executed).

`/* It is multi line comment.`

`It will not be displayed */`

```
document.write("example of javascript mult  
iline comment");
```